

Qualifying Exam in Microeconomics
Arizona State University
August 2017

There are four questions and you have four hours to complete the exam. If you think a question is in error, please clearly explain the error and continue working.

1. Let $\mathcal{E}$ be a private-ownership Arrow-Debreu economy with L goods indexed by $\ell = 1, \dots, L$, I consumers indexed by $i = 1, \dots, I$, and J firms, indexed by $j = 1, \dots, J$. All consumers have strongly monotone preferences ($x_i \geq x'_i, x_i \neq x'_i \implies x_i \succ_i x'_i$) on the consumption set $X_i = \mathbb{R}_+^L$. Let (p, x, y, T) be given, where p is a vector of prices, $T = (T_1, \dots, T_J)$ specifies an income transfer for each consumer, $x = (x_1, \dots, x_I)$ is a consumption assignment such that $\forall i, x_i \in D_i(p, y, T)$, and $y = (y_1, \dots, y_J)$ is a production plan. Assume the first $L - 1$ markets clear and prove, in detail, that the L market also clears.
- ask
micro
ok*

2. A consumer's preference relation $\succsim$ on $X \times X$ with $X = \mathbb{R}_+^L$ is *complete*, *transitive*, and *locally nonsatiated*.

- (a) Prove that if $x \succsim y$ and $y \succ z$, then $x \succ z$. *use def!*
- (b) Would the conclusion from part (a) hold if completeness of $\succsim$ were dropped? If it would fail, then give a counterexample or explain why the conclusion would fail. If it would hold, and your proof in part (a) used completeness, prove it without completeness (otherwise just point out you did not use completeness).
- (c) Suppose that x^* is in the consumer's demand correspondence at price-wealth pair (p, w) , and that $x' \succsim x^*$. Prove that $p \cdot x' \geq p \cdot x^*$. (Suggestion: use part (a).)
- (d) What is the role does the conclusion of part (c) play in proving the First Welfare Theorem? A sentence or two will suffice; we are *not* asking you to prove the First Welfare Theorem.

3. Consider a standard model of market signaling. After observing her type t , a worker chooses a level of education $e \geq 0$. The market, i.e. the second player, observes e and offers a wage w . The worker accepts or rejects the offer.

The worker's private information t is the worker's productivity. It is common knowledge that t is distributed uniformly in $[0, 1]$. The cost to acquire education level e for a worker of type t is e/t . The utility of a worker of type t who chooses education e and accepts a wage w is $U(t, e, w) = w - \frac{e}{t}$. The utility of a worker of type t who chooses education e and does not accept a wage offer is $-\frac{e}{t}$.

Education has no productive value for firms. The assumption of a competitive market is interpreted as a zero profit condition, that is to say, the wage offered given the available information to the market, is the expected productivity given the available information to the market.

- (a) Write down the strategy set of a worker.
- (b) For each $n = 1, 2, 3, \dots$ construct a "partial pooling" equilibrium in which, $\forall k, 1 \leq k \leq n$, all workers with type in the interval $[\frac{k-1}{n}, \frac{k}{n}]$ obtain the same education $e(k, n)$.
- (c) Identify the type of workers that benefit from moving from the equilibrium in the previous point to an equilibrium with $2n$ levels.
- (d) For the case $n = 2^k$, determine whether the total cost of education increases or decreases with changes in k .

other
letter

1.0
0.25

0.5
0.5

4. A monopoly firm produces and sells a single physical good at two dates. The date- t cost function for $t = 1, 2$ is $c_t(q_t) = kq_t$, where $k > 0$. The inverse demand for the good at date 1 is $P_1(q_1)$. The inverse demand for the good at date 2 depends on the consumption at date 1; it is $P_2(q_2, q_1)$, where, for each $q_2 \geq 0$, $P_2(q_2, \cdot)$ is a *strictly increasing* function of q_1 : in words, the higher the consumption at date 1, the higher the demand at date 2.

In addition $P_1(x) = P_2(x, 0)$ for every $x \geq 0$: if the firm sells nothing at date 1, the date-2 demand for the good is the same as the date-1 demand. The inverse demand at each date t is positive and *strictly decreasing* in q_t on $\mathbb{R}_+$.

The firm *does not discount* date-2 profit. Let (q_1^*, q_2^*) be any maximizer of the sum of the firm's profit across the two dates on $\mathbb{R}_+^2$. (Note that the objective need not be concave or even quasiconcave in (q_1, q_2) .)

Define q_1^0 to be the output that maximizes *date-1* profit, $\pi_1(q_1) = P_1(q_1)q_1 - kq_1$, on $\mathbb{R}_+$.

(a) Define

$$V(q_1) = \max_{q_2 \geq 0} [P_2(q_2, q_1)q_2 - kq_2],$$

the maximum date-2 profit if the firm produces q_1 at date 1. Prove that $V(\cdot)$ is *nondecreasing*. Why in general can you not strengthen *nondecreasing* to *strictly increasing*?

- (b) Prove that $q_1^* \geq q_1^0$. We suggest you start with the $q_1^0 = 0$ case. For the $q_1^0 > 0$ case, we suggest you proceed in the following steps. If you have trouble with an earlier step, then *assume* its conclusion in proving later steps. In particular we suggest that you *begin* the $q_1^0 > 0$ case by explaining how iv. follows from ii. and iii.

- i. For any $q_1 > 0$, $V(q_1) > 0$;
- ii. V is *strictly increasing* on $\mathbb{R}_{++}$;
- iii. $q_1^* > 0$; and finally $\rightarrow$ *weird!*
- iv. $q_1^* \geq q_1^0$.